



## FURTHER MATHEMATICS UNIT 3 APPLICATION TASK SAC 1 2019

### CORE – DATA ANALYSIS : BUYING A CAR

**VCAA NUMBER:** \_\_\_\_\_ **TEACHER:** \_\_\_\_\_

**Dates:** Wednesday 24<sup>th</sup> April 3:30-5:40PM (all students)

Plus 3 periods in class (1 double and 1 single) in the following few days: Friday 26<sup>th</sup> April to Tuesday 30<sup>th</sup> April

**Duration:** 10 minutes reading time plus 2 hours plus 3 x 50 minutes.

**Materials allowed:** 1 x Bound reference, CAS calculator, scientific calculator, ruler, pens/pencils

#### Outline of SAC

Students will be given PART ONE of the SAC on Wednesday 24<sup>th</sup> April and will have 10 minutes reading time plus 2 hours on Monday 23<sup>rd</sup> April. They will be given PART TWO to complete in a single lesson on Friday 26<sup>th</sup> April and PART THREE to complete in a double lesson Monday 29<sup>th</sup> or Tuesday 30<sup>th</sup> April.

At the end of each session the student must:

1. Submit their SAC,
2. Submit their bound reference,
3. Submit their CAS calculator
4. Sign to state that they have submitted all these items and that the work is their own.

Students will be given their SAC, bound reference and CAS back at the start of each session.

#### **Total Marks: 83**

##### **PART ONE**

Question 1 : 16mks      Question 2 : 13mks

##### **PART TWO**

Question 3 : 17mks

##### **PART THREE**

Question 4 : 17mks      Question 5: 20mks

**In order to satisfactorily complete this SAC** students must demonstrate an understanding of the following three outcomes. These outcomes are covered throughout the SAC.

The following table summarises what each outcome entails in terms of key knowledge and skills. Please note that not all points are covered in this SAC- indicated by an NA (not assessed).

| <b>Outcome 1 (10 marks):</b> On completion of this unit the student should be able to define and explain key concepts and apply related mathematical techniques and models as specified in Area of Study 1 in routine contexts. To achieve this outcome the student will draw on knowledge and skills outlined in Area of Study 1. |              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Key knowledge</b>                                                                                                                                                                                                                                                                                                               | Satisfactory |
| • types of data: categorical (nominal and ordinal) and numerical (discrete and continuous)                                                                                                                                                                                                                                         |              |
| • frequency tables, bar charts including segmented barcharts, histograms, stem plots, dot plots, and their application in the context of displaying and describing distributions                                                                                                                                                   |              |
| • log (base 10) scales, and their purpose and application                                                                                                                                                                                                                                                                          |              |
| • five-number summary and boxplots (including the designation and display of possible outliers)                                                                                                                                                                                                                                    |              |
| • mean and standard deviation                                                                                                                                                                                                                                                                                                      |              |
| • normal model and the 68–95–99.7% rule, and standardised values (z-scores)                                                                                                                                                                                                                                                        |              |
| • response and explanatory variables                                                                                                                                                                                                                                                                                               |              |
| • two-way frequency tables, segmented bar charts, back-to-back stem plots, parallel boxplots, and scatterplots, and their application in the context of identifying and describing associations                                                                                                                                    |              |
| • correlation coefficient, $r$ , its interpretation, the issue of correlation and cause and effect                                                                                                                                                                                                                                 |              |
| • least squares line and its use in modelling linear associations                                                                                                                                                                                                                                                                  |              |
| • data transformation and its purpose                                                                                                                                                                                                                                                                                              |              |
| • time series data and its analysis                                                                                                                                                                                                                                                                                                | NA           |
| <b>Key skills</b>                                                                                                                                                                                                                                                                                                                  |              |
| • construct frequency tables and bar charts and use them to describe and interpret the distributions of categorical variables                                                                                                                                                                                                      |              |
| • answer statistical questions that require a knowledge of the distribution/s of one or more categorical variables                                                                                                                                                                                                                 |              |
| • construct stem and dot plots, boxplots, histograms and appropriate summary statistics and use them to describe and interpret the distributions of numerical variables                                                                                                                                                            |              |
| • answer statistical questions that require a knowledge of the distribution/s of one or more numerical variables                                                                                                                                                                                                                   |              |
| • solve problems using the z-scores and the 68–95–99.7% rule                                                                                                                                                                                                                                                                       |              |
| • construct two-way tables and use them to identify and describe associations between two categorical variables                                                                                                                                                                                                                    | NA           |
| • construct parallel boxplots and use them to identify and describe associations between a numerical variable and a categorical variable                                                                                                                                                                                           |              |
| • construct scatterplots and use them to identify and describe associations between two numerical variables                                                                                                                                                                                                                        |              |
| • calculate the correlation coefficient, $r$ , and interpret it in the context of the data                                                                                                                                                                                                                                         |              |
| • answer statistical questions that require a knowledge of the associations between pairs of variables                                                                                                                                                                                                                             |              |
| • determine the equation of the least squares line giving the coefficients correct to a required number of decimal places or significant figures as specified                                                                                                                                                                      |              |
| • distinguish between correlation and causation                                                                                                                                                                                                                                                                                    |              |
| • use the least squares line of best fit to model and analyse the linear association between two numerical variables and interpret the model in the context of the association being modelled                                                                                                                                      |              |
| • calculate the coefficient of determination, $r^2$ , and interpret in the context of the association being modelled and use the model to make predictions, being aware of the problem of extrapolation                                                                                                                            |              |
| • construct a residual analysis to test the assumption of linearity and, in the case of clear non-linearity, transform the data to achieve linearity and repeat the modelling process using the transformed data                                                                                                                   |              |
| • identify key qualitative features of a time series plot including trend (using smoothing if necessary), seasonality, irregular fluctuations and outliers, and interpret these in the context of the data                                                                                                                         | NA           |
| • calculate, interpret and apply seasonal indices                                                                                                                                                                                                                                                                                  | NA           |
| • model linear trends using the least squares line of best fit, interpret the model in the context of the trend being modelled, use the model to make forecasts being aware of the limitations of extending forecasts too far into the future                                                                                      |              |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |           |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|
| <b>Outcome 2 (20 marks):</b> On completion of this unit the student should be able to select and apply the mathematical concepts, models and techniques as specified in Area of Study 1 in a range of contexts of increasing complexity. To achieve this outcome the student will draw on knowledge and skills outlined in Area of Study 1                                                                                                                                     |           |           |           |
| <b>Key knowledge</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |           |           |           |
| <ul style="list-style-type: none"> <li>the facts, concepts and techniques associated with data analysis</li> </ul>                                                                                                                                                                                                                                                                                                                                                             |           |           |           |
| <ul style="list-style-type: none"> <li>standard models studied in data analysis and their area of application</li> </ul>                                                                                                                                                                                                                                                                                                                                                       |           |           |           |
| <ul style="list-style-type: none"> <li>general formulation of the concepts, techniques and models studied in data analysis</li> </ul>                                                                                                                                                                                                                                                                                                                                          |           |           |           |
| <ul style="list-style-type: none"> <li>assumptions and conditions underlying the use of the concepts, techniques, and models associated with data analysis</li> </ul>                                                                                                                                                                                                                                                                                                          |           |           |           |
| <b>Key skills</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |           |           |
| <ul style="list-style-type: none"> <li>identify, recall and select facts, concepts, models and techniques needed to investigate and analyse statistical features of a data set with several variables that can include time series data</li> </ul>                                                                                                                                                                                                                             |           |           |           |
| <ul style="list-style-type: none"> <li>interpret and report the results of a statistical investigation or of completing a modelling or problem-solving task in terms of the context under consideration, including discussing the assumptions in application of these models</li> </ul>                                                                                                                                                                                        |           |           |           |
| <b>Outcome 3 (10 marks):</b> On completion of this unit the student should be able to select and appropriately use numerical, graphical, symbolic and statistical functionalities of technology to develop mathematical ideas, produce results and carry out analysis in situations requiring problem-solving, modelling or investigative techniques or approaches. To achieve this outcome the student will draw on knowledge and related skills outlined in Area of Study 1. |           |           |           |
| <b>Key knowledge</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |           |           |           |
| <ul style="list-style-type: none"> <li>the difference between exact numerical and approximate numerical answers when using technology to perform</li> </ul>                                                                                                                                                                                                                                                                                                                    |           |           |           |
| <ul style="list-style-type: none"> <li>computation, and rounding to a given number of decimal places or significant figures</li> </ul>                                                                                                                                                                                                                                                                                                                                         |           |           |           |
| <ul style="list-style-type: none"> <li>domain and range requirements for specification of graphs of models and relations, when using technology</li> </ul>                                                                                                                                                                                                                                                                                                                     |           |           |           |
| <ul style="list-style-type: none"> <li>the relation between numerical, graphical and symbolic forms of information about models and equations and</li> </ul>                                                                                                                                                                                                                                                                                                                   |           |           |           |
| <ul style="list-style-type: none"> <li>the corresponding features of those functions and equations</li> </ul>                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |
| <ul style="list-style-type: none"> <li>similarities and differences between formal mathematical expressions and their representation by technology</li> </ul>                                                                                                                                                                                                                                                                                                                  |           |           |           |
| <ul style="list-style-type: none"> <li>the selection of an appropriate functionality of technology in a variety of mathematical contexts</li> </ul>                                                                                                                                                                                                                                                                                                                            |           |           |           |
| <b>Key skills</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |           |           |
| <ul style="list-style-type: none"> <li>distinguish between exact and approximate presentations of mathematical results produced by technology, and interpret these results to a specified degree of accuracy in terms of a given number of decimal places or significant figures</li> </ul>                                                                                                                                                                                    |           |           |           |
| <ul style="list-style-type: none"> <li>use technology to carry out numerical, graphical and symbolic computation as applicable</li> </ul>                                                                                                                                                                                                                                                                                                                                      |           |           |           |
| <ul style="list-style-type: none"> <li>produce tables of values, families of graphs and collections of other results using technology, which support general analysis in problem-solving, investigative and modelling contexts</li> </ul>                                                                                                                                                                                                                                      |           |           |           |
| <ul style="list-style-type: none"> <li>identify the relation between numerical, graphical and symbolic forms of information about models and equations and the corresponding features of those models and equations</li> </ul>                                                                                                                                                                                                                                                 |           |           |           |
| <ul style="list-style-type: none"> <li>select an appropriate functionality of technology in a variety of mathematical contexts, related to data analysis, and provide a rationale for these selections</li> </ul>                                                                                                                                                                                                                                                              |           |           |           |
| <ul style="list-style-type: none"> <li>relate the results from a particular technology application to the nature of a particular mathematical task (investigative, problem solving or modelling) and verify these results</li> </ul>                                                                                                                                                                                                                                           |           |           |           |
| <ul style="list-style-type: none"> <li>specify the process used to develop a solution to a problem using technology, and communicate the key stages of mathematical reasoning (formulation, solution, interpretation) used in this process</li> </ul>                                                                                                                                                                                                                          |           |           |           |
| Overall result:                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Outcome 1 | Outcome 2 | Outcome 3 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |           |           |

After SACs have been marked, teachers will decide if a student has demonstrated enough knowledge throughout their SAC to be given a satisfactory. If a student can't demonstrate through her SAC that she understands the content covered then the student must demonstrate her knowledge by either: 1. Showing her teacher her up-to-date workbook; and/or, 2. Completing a short test.

## SAC

### INTRODUCTION:

New cars are safer to drive, but they are expensive to buy. This SAC explores the prices of cars of various ages, various sizes and various makes and attempts to find a satisfactory relationship between the age of a car and its price.

Ms G would like her son Rourke to buy a recent model car because of the additional safety features. She decides to investigate 2016 models of cars on [carsales.com.au](https://carsales.com.au) and collates the average price of cars that are for sale in March. All vehicles are standard automatic petrol sedans.

Ms G is interested in a few different research questions.

1. Is there a difference between sedan size and reaction price?
2. Do some cars lose value over time more quickly than others?
3. Is it possible to predict a car's price from its age?

These questions and others will be investigated in this assessment.

## **PART ONE**

### **2016 Car Prices**

| <b>Car Prices</b> |                   |                        |
|-------------------|-------------------|------------------------|
| <b>Car Type</b>   | <b>Sedan Size</b> | <b>2016 Price (\$)</b> |
| Ford Focus        | Small             | 18 308                 |
| Honda Civic       | Small             | 20 216                 |
| Holden Cruze      | Small             | 15 530                 |
| Hyundai Elantra   | Small             | 18 090                 |
| Kia Cerato        | Small             | 15 674                 |
| Mazda 3           | Small             | 19 882                 |
| Maserati Ghibli   | Small             | 95 518                 |
| Mitsubishi Lancer | Small             | 16 302                 |
| Nissan Pulsar     | Small             | 14 730                 |
| Subaru Impreza    | Small             | 18 499                 |
| Toyota Corolla    | Small             | 16 702                 |
| Audi A4           | Medium            | 55 282                 |
| BMW M3            | Medium            | 103 043                |
| Holden Malibu     | Medium            | 14 310                 |
| Honda Accord      | Medium            | 29 022                 |
| Hyundai Sonata    | Medium            | 18 800                 |
| Jaguar XE         | Medium            | 56 504                 |
| Kia Optima        | Medium            | 22 367                 |
| Mazda 6           | Medium            | 29 071                 |
| Mercedes Benz C63 | Medium            | 108 813                |
| Nissan Altima     | Medium            | 17 404                 |
| Subaru Liberty    | Medium            | 24 470                 |
| Toyota Camry      | Medium            | 19 709                 |
| Toyota Prius      | Medium            | 24 554                 |



**Question 1 : Analysing the 2016 data. (16 marks)**

- a. State whether the following variables are numerical (discrete or continuous) or categorical (nominal or ordinal). Two words are needed to describe each variable.

i. Sedan Size

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ii. Car Type

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iii. Price

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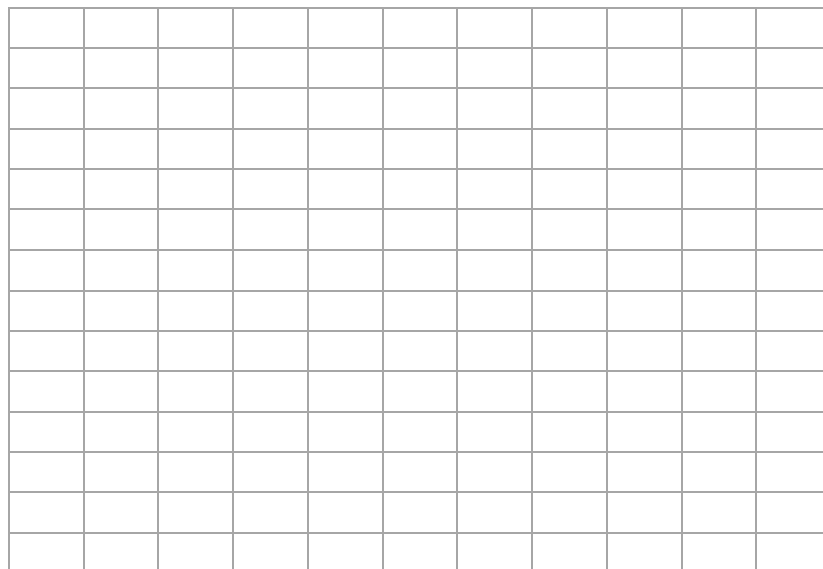
(3 marks)

- b. Complete the frequency table below for 2016 car prices for all cars.

| Price                | Frequency | Interval               | Frequency |
|----------------------|-----------|------------------------|-----------|
| \$10 000- < \$20 000 |           | \$60 000- < \$70 000   |           |
| \$20 000- < \$30 000 |           | \$70 000- < \$80 000   |           |
| \$30 000- < \$40 000 |           | \$80 000- < \$90 000   |           |
| \$40 000- < \$50 000 |           | \$90 000- < \$100 000  |           |
| \$50 000- < \$60 000 |           | \$100 000- < \$110 000 |           |

(2 marks)

- c. Construct an appropriate graph on the grid below to summarize the 2016 car data.



(3 marks)

- d.** Using your graph from part **c.**, describe the 2016 Car Price data in terms of shape, centre and spread.

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(3 marks)

- e.** Ms G realised that she may have included Luxury cars in her search. Perform an outlier test and state the Luxury cars in the 2016 data.

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(3 marks)

- f.** Ms G would like to give Rourke an idea of the average amount he should save to buy a car. Should she use the original data? Explain.

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(2 marks)



**Question 2 : Refining the search. (13 marks)**

- a. Ms G decides to take out the luxury cars from her 2016 search and investigate the difference in price of small sedans compared to medium sedans. Calculate the five number summaries (to the nearest dollar) for the prices of cars for each of the sedan sizes.

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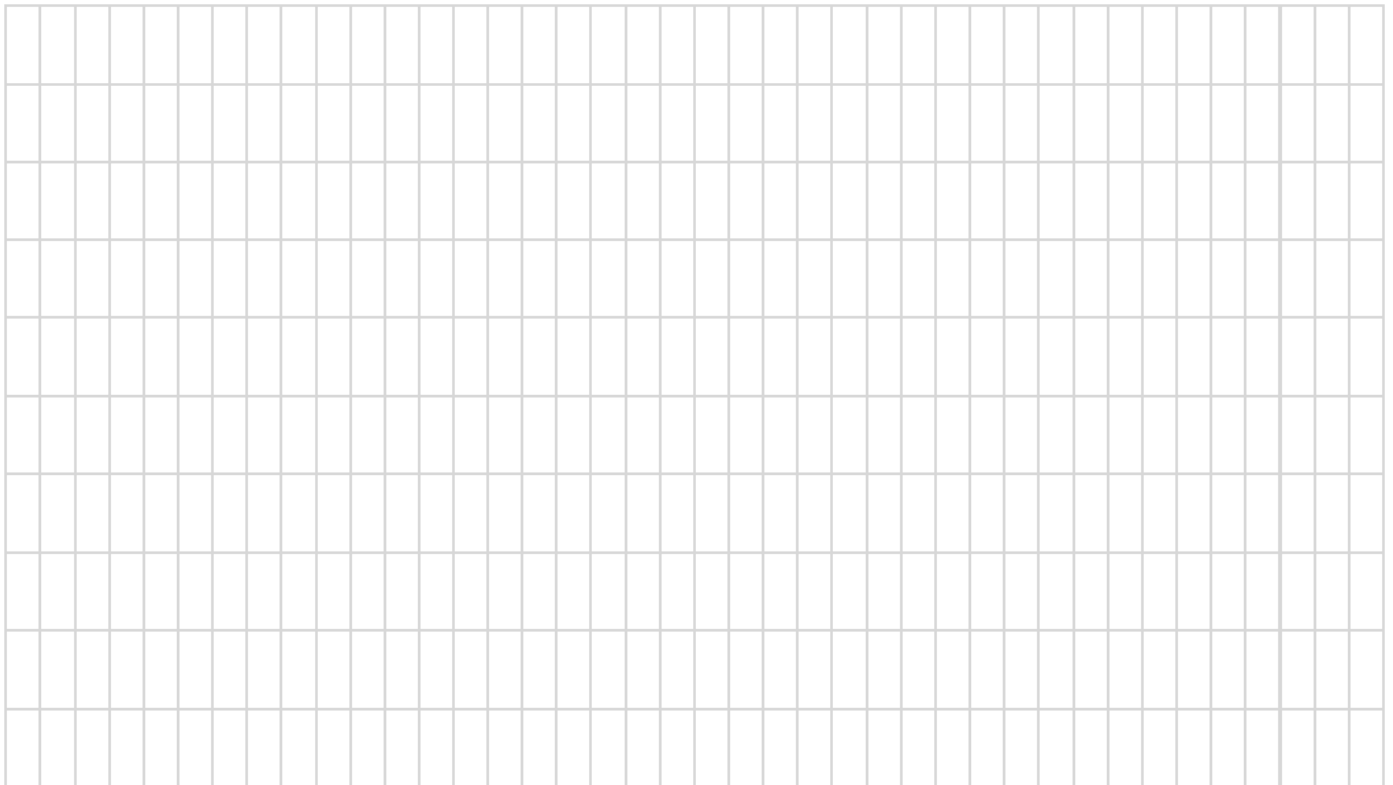
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(2 marks)

- b. Construct parallel boxplots for the data for 2016 small and medium non-luxury sedans.



(4 marks)

- c.** Comparing the shape, centre and spread of the boxplots, compose a summary using statistics for Ms G to give to Rourke regarding the price of 2016 small non-luxury sedans compared to medium non-luxury sedans.

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(3 marks)

Using the parallel boxplots you constructed, answer question d.

- d.** Approximately what percentage of 2016 small sedans and medium sedans were above \$18 000?

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(2 marks)

- e.** Give two possible explanations for the differences in the 2016 small and medium sedan box plots.

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(2 marks)

## **PART TWO**

VCAA NUMBER: \_\_\_\_\_

TEACHER: \_\_\_\_\_

### **Question 3 : Which cars hold their value the most over time? (17 marks)**

Rourke has decided he can only afford a small sedan. Ms G wants him to make a good investment so she was wondering if some cars decreased in value faster than others. She explored the current 2019 prices of the same vehicles and this data can be found in table 2. She omitted the luxury cars and the Holden and Nissan were not produced in 2019.

| <b>Car Type</b>   | <b>2016 Price (\$)</b> | <b>2019 Price (\$)</b> |
|-------------------|------------------------|------------------------|
| Ford Focus        | 18 308                 | 24 390                 |
| Honda Civic       | 20 216                 | 24 390                 |
| Hyundai Elantra   | 18 090                 | 23 790                 |
| Kia Cerato        | 15 674                 | 22 290                 |
| Mazda 3           | 19 882                 | 25 490                 |
| Mitsubishi Lancer | 16 302                 | 21 000                 |
| Subaru Impreza    | 18 499                 | 22 890                 |
| Toyota Corolla    | 16 702                 | 24 370                 |

- a. Calculate the mean and standard deviation for the non-luxury 2016 data and for the 2019 data found in the table above. Place the values in the table below to 2 significant figures.

|                           | <b>2016</b> | <b>2019</b> |
|---------------------------|-------------|-------------|
| <b>Mean</b>               |             |             |
| <b>Standard Deviation</b> |             |             |

(2 marks)

- b. Is the mean an appropriate measure of centre for these car prices? Explain why or why not.

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(2 marks)

- c. Ms G is going to use the 68-95-99.7% rule to analyse the 2016/2019 data. Explain why the 68-95-99.7% rule should or should not be used with this data.

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(1 mark)

- d. Using CAS, randomly select one of the cars from the above 2016/2019 data. State below the car and the method that you used on CAS to select this vehicle.

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(1 mark)

- e. Using your results from a., calculate the standardised score for your selected vehicle for the 2019 data correct to 2 decimal places.

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(2 marks)

- f. For the standardised score obtained in part e. above, determine the approximate percentage (or range) for the number of cars below this value and explain your reasoning.

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(2 marks)

- g. Using your results from a., calculate the standardised score for the same vehicle for the 2016 car data correct to 2 decimal places.

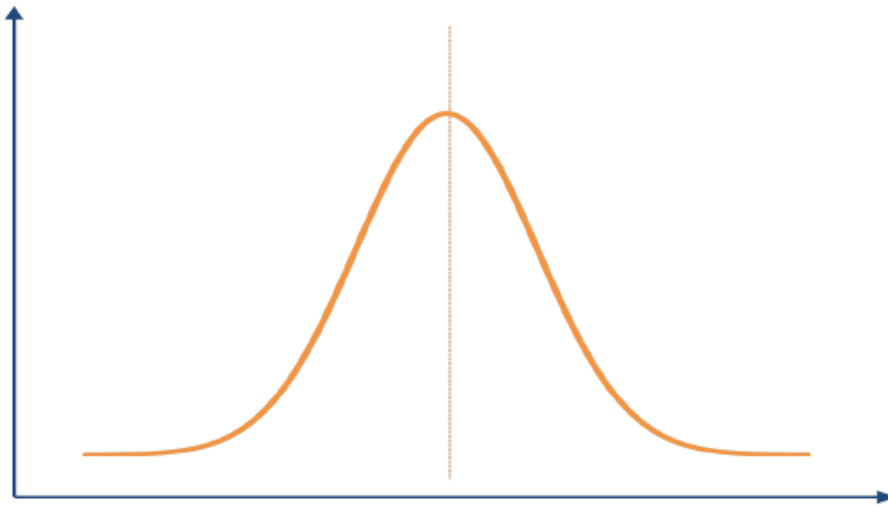
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(2 marks)

- h.** On the normal graph below, approximately place your values found in parts **e.** and **g.** and label them. Compare and comment on what the 2 standard scores imply for your chosen car.



(2 marks)

- i.** Rourke is thinking of buying a car in the top 16% price range. Using your results from **a.**, determine the minimum amount that he would spend on a 2016 car and a 2019 car.

(2 marks)

- j.** What is the actual percentage of 2016 cars that fall in the top 16% price range?

(1 mark)

## **PART THREE**

VCAA NUMBER: \_\_\_\_\_

TEACHER: \_\_\_\_\_

### **Question 4 : How about a Mazda 3? (17 marks)**

Rourke decides that he really likes the Mazda 3. Ms G is interested in analysing the relationship between the age of a Mazda 3 sedan and its price.

| Mazda 3 Car Prices |            |     |
|--------------------|------------|-----|
| Year               | Price (\$) | Age |
| 2019               | 25 490     | 1   |
| 2018               | 21 407     | 2   |
| 2017               | 19 258     | 3   |
| 2016               | 17 572     | 4   |
| 2015               | 15 942     | 5   |
| 2014               | 14 660     | 6   |
| 2013               | 12 890     | 7   |
| 2012               | 11 725     | 8   |
| 2011               | 11 230     | 9   |
| 2010               | 9 526      | 10  |
| 2009               | 8 733      | 11  |
| 2008               | 6 062      | 12  |
| 2007               | 7 660      | 13  |
| 2006               | 6 332      | 14  |
| 2005               | 6 180      | 15  |
| 2004               | 4 925      | 16  |

- a.** In this case, what is the response variable?

\_\_\_\_\_

(1 mark)

- b.** Use CAS to plot the above points and describe the relationship between the age of the Mazda 3 and the price in dollars in terms of strength, direction and form.

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\_\_\_\_\_

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(3 marks)

- c. The value of the coefficient of determination for the least squares relationship between the price and the age is 0.9397, correct to 4 decimal places. Interpret the meaning of this value in terms of the given data.

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(1 mark)

- d. Using the information given in part c. above, determine the value of Pearson's correlation coefficient for this data correct to 4 decimal places.

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(1 mark)

- e. Given the following information and also using your answer to part d., calculate the slope and the y-intercept for the equation of the least squares regression line for this data. Show your working. Then write the equation using correct variables and the coefficients correct to 3 significant figures.

**mean age of car = 8.5**

**mean price of car = \$12 500**

**standard deviation of car ages = 4.76**

**standard deviation of car prices = \$6 100**

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(3 marks)

- f. Explain the meaning in real terms of the slope of this line.

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(1 mark)

- g. Use the least squares regression equation to calculate the price of a 5-year-old Mazda 3.

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(1 mark)

- h. Rourke is looking at a 5 year old Mazda 3 which is advertised at \$14 999. Calculate the residual of the price for this vehicle. Explain what this means to Rourke.

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(2 marks)

- i. Use CAS to perform a linear regression and check your answer to e. is close. Complete the following table of residuals correct to the nearest dollar for the Mazda 3 data.

| Age of Mazda 3 | Residual of price (\$) |
|----------------|------------------------|
| 1              | 3 671                  |
| 2              |                        |
| 3              | -69                    |
| 4              |                        |
| 5              | -893                   |
| 6              |                        |
| 7              | -1453                  |
| 8              |                        |
| 9              | -622                   |
| 10             | -1 080                 |
| 11             | -627                   |
| 12             | -2 052                 |
| 13             | 792                    |
| 14             | 710                    |
| 15             | 1 804                  |
| 16             | 1 795                  |

(2 marks)

- j. Use CAS to see the residual plot. What does the residual plot indicate about the relationship between age of a Mazda 3 and its price?

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(2 marks)



**Question 5 : Maybe a transformation is needed? (20 marks)**

Ms G has come to the conclusion that a transformation may enable her to make a more accurate prediction of the price of the Mazda 3 given its age. Use the table at the start of Question 4 to help you answer the following questions.

- a.** Perform each of the transformations listed below and write the value of  $r^2$  and the coefficients in the equations correct to 2 significant figures. Describe the residual plot of each transformation.

| Relationship                 | $r^2$ value | Equation | Residual plot description |
|------------------------------|-------------|----------|---------------------------|
| Price vs. age                |             |          |                           |
| Price vs. $\log(\text{age})$ |             |          |                           |
| Price vs. $1/\text{age}$     |             |          |                           |
| $\log(\text{price})$ vs. age |             |          |                           |
| $1/\text{price}$ vs. age     |             |          |                           |

(5 marks)

- b.** State which relationship you consider best and give reasons for your answer.

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(2 marks)

- c.** Using your equation from part **a.** above, predict the price of a 5-year-old Mazda 3 using the relationship that you consider best correct to the nearest dollar.

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(2 marks)

- d.** Comment on the reliability of your prediction in part **c.**

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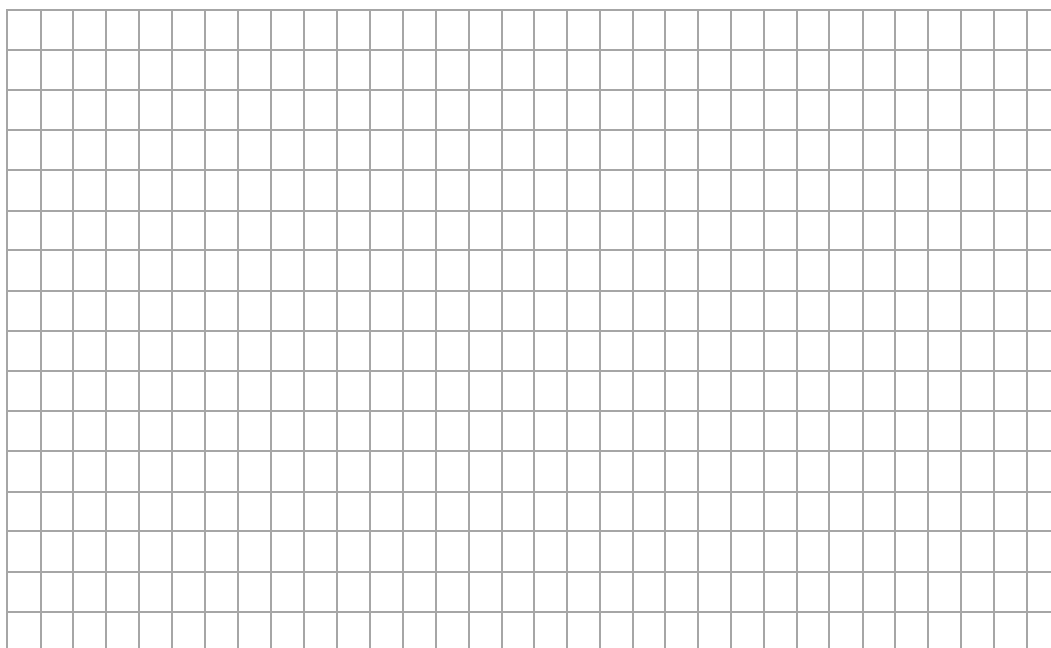
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(2 marks)

- e. Construct a graph of the relationship that you consider best, showing the scatterplot of the relationship **after** transformation.



(3 marks)

- f. Does the scatterplot after transformation indicate that the transformation was successful? Explain your answer.

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(2 marks)

- g. Calculate the residual for the 5 year old Mazda 3 advertised at \$14 999.

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(2 marks)

- h. Compare the residual you obtained in g. above to the residual you obtained in **Question 4** part h. Explain, referring to the result of the transformation, **why** any difference was observed.

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(2 marks)

**END OF SAC**